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Leveraging Explainable Machine Learning Algorithms and Social Network Analysis for Detecting Money Laundering Patterns in Financial Transactions

by

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List of Acronyms

AIS FinCEN Artificial Intelligence System

AML/CTF Anti Money Laundering and Counter terrorist financing

BOU Bank Of Uganda

CTR Currency Transaction Report

ESAAMLG Eastern and Southern Africa Anti-Money Laundering Group

FIA Financial Intelligence Authority

GDP Gross Domestic Product

GDPR General Data Protection Regulation

IFFs Illicit Financial Flows

KNN K-Nearest Neighbors

LIME Local Interpretable Model-agnostic Explanations

ML Machine Learning

SHAP SHapley Additive exPlanations

SNA Social Network Analysis

SSA Sub Saharan Africa

STR Suspicious Transaction Report

SWIFT Society for Worldwide Interbank Financial Telecommunications

UNISS Uganda National Inter-bank Settlement System

1 Introduction

1.1 Background

Money laundering is a global problem that affects the financial stability and security of many regions, especially Sub-Saharan Africa. Money laundering is the process of concealing the origin, ownership, and destination of money obtained from illegal sources, such as corruption, terrorism, drug trafficking, tax evasion, and other crimes [1]. Money laundering can facilitate the financing of criminal activities, distort the allocation of resources, and erode the rule of law and the development of the region.

Sub-Saharan Africa is particularly vulnerable to money laundering due to its large informal economy, weak regulatory and institutional frameworks, high levels of corruption and poverty, and exposure to cross-border and transnational crimes. According to the African Development Bank and Global Financial Integrity report, it is estimated that Africa lost between \$1.3 and \$1.4 trillion in illicit financial flows between 1980 and 2009, which is equivalent to 54.5% of its gross domestic product (GDP) [2] [3]. According to a policy brief by the Brookings Institution, it estimates that around \$50 to \$60 billion per year leaves Africa in the form of illicit flows posing a challenge to development financing [4]. These illicit flows include money laundering, tax evasion, trade mis-invoicing, and other forms of financial crime.

According to a United Nations report, Illicit financial flows (IFFs) in Africa hinder progress and prevent sustainable development [5]. The main sources of IFFs in Africa are commercial practices, criminal activities, trafficking, smuggling, and corruption. The process of money laundering is used for many purposes such as sabotage of a country's economic condition, financing illegal activities, and legitimizing illegal funds. The growth of internet technology (cyber payments and internet money transfers) and the loosely coupled nature of fund transfer gateways (RTGS and SWIFT) have led to the increase of money laundering activities within the African continent.

The total amount laundered internally and brought into the Eastern and Southern Africa Anti-Money Laundering Group(ESAAMLG) region during 1999 was equal to just over \$1,807

Country	Internal	Outgoing	Total Gen-erated	Incoming	Total Laundered	% Internal	% Outgoing
Botswana	123.4	117.6	241.0	1722	1845.0	51.2	48.8
Kenya	73.8	74.0	147.8	777	811.5	84.0	16.0
Lesotho	0.9	0.1	1.4	458	4587	86.4	13.6
Malawi	96.4	47.0	143.4	1587	1732	67.2	32.8
Mauritius	95.2	61.4	156.6	1129	1224.5	60.8	39.2
Mozambique	87.6	34.1	121.7	17	104.8	72.0	28.0
Namibia	114.6	38.3	152.9	489	603.5	57.6	42.4
Seychelles	68.1	4.8	73.0	1963	2031.5	93.4	6.6
South Africa	6143.7	4095.8	10239.5	566	7099.9	60.00	40.00
Amounts in Millions of USD							

Table 1: Kiwanuka, P. I(2020),Estimates of money Laundering Activity.

billion. It is conceivable that some of the internally laundered proceeds were subsequently externalized, and that some of the proceeds introduced into the region were taken out again within the year [6].

According to Kiwanuka, P. I(2020), the data in Table 1, approximates that over \$1 trillion is laundered within some African countries and this creates a big problem. The estimated amount of money laundered globally in one year is 2 - 5 percent of global GDP, or \$800 billion - \$2 trillion in current US dollars according to the United Nations [7].

Many approaches have been discussed and developed for the detection of money laundering such as internal policies within institutions. Most of them have the following shortcomings namely, identifying the root of money laundering activities, detecting transactions below defined thresholds, and a high number of false positives [8].

This research aims to develop a comprehensive approach to detecting money laundering patterns using machine learning algorithms and advanced data analysis techniques. The continent's diverse cultures, political systems, and economic landscapes make it a breeding ground for illicit cash flows.

1.2 Problem Statement

Money laundering poses a growing threat to the economies and stability of Sub-Saharan Africa, exacerbated by factors such as informal economies, cash-based transactions, weak regulatory frameworks and vulnerability to corruption.

While Explainable Artificial Intelligence (XAI) and Social Network Analysis (SNA) have individually shown promise in enhancing AML detection, both approaches possess inherent weaknesses that limit their standalone effectiveness. XAI models, particularly post-hoc explainability tools such as LIME and SHAP, can suffer from instability in explanations, vulnerability to adversarial manipulation, and a lack of consensus on standardized interpretability metrics. In high-stakes financial environments, this can result in explanations that are either too abstract for investigators or too granular to be operationally useful.

Similarly, SNA faces challenges in terms of scalability and data quality. Large transaction networks are computationally expensive to process, especially in real time. Incomplete or noisy network data can lead to misleading centrality measures or incorrect detection of communities, reducing trust in the system’s outputs. Additionally, network-based features can be susceptible to overfitting when historical laundering structures do not generalize to emerging criminal strategies.

Traditional rule-based AML systems further exacerbate the problem by being inflexible and easily circumvented by structured layering and smurfing techniques. As a result, the region continues to suffer from undetected illicit flows despite compliance frameworks being in place.

This research addresses these weaknesses by integrating XAI with SNA into a hybrid supervised learning framework. While unsupervised learning has been recommended for discovering entirely unknown patterns, the availability of labeled transactional data from Interswitch and the regulatory requirement for transparent, auditable decisions make supervised learning the most appropriate choice for this context.

1.3 Objectives of the Study

1.3.1 Main Objective

The main objective of the study is to provide empirical evidence on the effectiveness of combining social network analysis (SNA) and Explainable Machine Learning for Anti Money Laundering detection.

1.3.2 Specific Objectives

This research will especially

- (i) Identify and characterize properties of common money laundering scenarios.
- (ii) Identify and select the most relevant features and data sources for training the explainable machine learning model and social network analysis for money laundering detection.
- (iii) Develop and evaluate an explainable machine learning model with network analysis capable of identifying anomalous patterns and potential money laundering activities in Sub-Saharan Africa.

1.4 Significance of the Study

The integration of machine learning algorithms in detecting money laundering patterns revolutionizes the financial industry's approach to combating illicit activities. This innovative methodology involves leveraging data analysis, pattern recognition, and adaptive models. Machine learning enables real-time monitoring, behavioral analysis, and anomaly detection, offering a nuanced and context-aware identification of suspicious transactions. Integrated with existing systems, machine learning reduces false positives, providing a more accurate and efficient means of detecting potential money laundering activities. This forward-looking strategy signifies a commitment to staying ahead of evolving threats and safeguarding the integrity of global financial systems.

2 Literature Review

2.1 Introduction

Ten countries in sub-Saharan Africa are grey-listed and are under increased monitoring by the Financial Action Task Force (FATF) for having strategic deficiencies in their AML/CTF frameworks. These countries are Botswana, the Democratic Republic of the Congo, Ethiopia, Ghana, Mauritius, Mozambique, Nigeria, Tanzania, Uganda, and Zimbabwe [9].

According to a report by the FATF, Sub-Saharan African Countries, like Uganda, have made some progress in combating money laundering since the country's Anti-Money Laundering Act was enacted in 2013 [10]. However, the police and other agencies such as the Financial Intelligence Authority (FIA) still face difficulties in enforcing the law [8]. This is because they lack enough resources to acquire advanced financial monitoring software platforms [11]. Without a system to track financial transactions, it is hard to fight money laundering. Many financial institutions in Uganda also struggle with the same problems. If the country does not improve its anti-money laundering regime, it could be isolated from the global financial system. This would affect the banking relationships between lenders and increase the cost of foreign loans for local businesses. These challenges are not only faced by Uganda, but they also cut across all the sub-Saharan countries.. [12].

2.2 Money Laundering

Money laundering is a criminal process in which illegally obtained funds, often stemming from illicit activities such as drug trafficking, corruption, or organized crime, are concealed through a series of complex transactions. The money laundering process typically involves three main stages: placement (introducing the illicit funds into the financial system), layering (creating complex transactions to distance the funds from their source), and integration (reintroducing the laundered funds into the legitimate economy) [13].

Detecting money laundering is a challenging task for both banks and regulators, as the perpetrators often use complex and sophisticated methods to evade the existing anti-money laundering (AML) rules and systems. Traditional AML systems rely on predefined rules

and thresholds to flag suspicious transactions based on certain criteria, such as the amount, frequency, location, and type of transactions. [14]

2.3 Money Laundering Patterns Hard to detect using Traditional methods

- (i) Payments to Cash-Based Businesses: Criminals funnel money through businesses that primarily deal in cash transactions, making it difficult to trace the source of funds. [8].
- (ii) False Invoices: Money launderers create fictitious invoices or manipulate legitimate invoices to move funds. Detecting these false transactions can be tricky. [8] [15].
- (iii) Small Amounts Below AML Threshold: Launderers split large sums into smaller transactions below the Anti-Money Laundering (AML) reporting threshold. These seemingly innocuous amounts can add up over time. [8].
- (iv) Trusts and Offshore Companies: Money launderers often use trusts and offshore companies to hide the true ownership of assets. These structures make it challenging to identify the ultimate beneficiaries. [8].
- (v) Foreign Bank Accounts: Criminals move funds across international borders, taking advantage of different regulations and jurisdictions to obscure the money trail. [8].

2.4 Limitations of Detecting Money Laundering Patterns using Traditional methods

Traditional AML systems have several limitations, such as:

They generate a high number of false positives, which increase the workload and cost of manual investigation and verification. They are unable to capture the dynamic and evolving nature of money laundering schemes, which often adapt to the changing regulations and technologies [16] [15].

They are vulnerable to manipulation and evasion by the money launderers, who can exploit the loopholes and weaknesses of the rules [15] .

To overcome these limitations, machine learning algorithms have emerged as a promising alternative or complement to the traditional rule-based systems. Machine learning is a branch of artificial intelligence that enables computers to learn from data and make predictions or decisions without explicit programming [17].

Machine learning algorithms can leverage the large and diverse data sources available in the financial sector [18], such as customer profiles, transaction histories, network graphs, and external information, to identify patterns and anomalies that indicate money laundering activities.

2.5 Machine learning algorithms applicable to the AML domain

Supervised learning: This type of machine learning requires labeled data, which means that each data point has a known outcome or class. For example, a supervised learning algorithm can be trained on a data set of transactions that are labeled as either normal or suspicious and then use the learned model to classify new transactions as such. Some of the common supervised learning algorithms used for AML are decision trees, random forests, logistic regression, support vector machines, and neural networks.

Unsupervised learning: This type of machine learning does not require labeled data, but rather tries to discover the underlying structure or patterns in the data. For example, an unsupervised learning algorithm can be used to cluster transactions based on their similarity or dissimilarity, and then detect outliers or anomalies that deviate from the normal clusters. Some of the common unsupervised learning algorithms used for AML are k-means, hierarchical clustering, principal component analysis, and autoencoders.

Reinforcement learning: This type of machine learning involves learning from trial and error, based on feedback or rewards received from the environment. For example, a reinforcement learning algorithm can be used to optimize AML policies or strategies, learning from the results of previous actions and choosing the best action for the current situation. Some of the common reinforcement learning algorithms used for AML are Q-learning, SARSA, and policy gradient.

2.5.1 Critical Analysis of Explainable AI Models

Recent studies have demonstrated that while SHAP and LIME are popular, their explanations can vary significantly across runs when the underlying model is retrained. For high-volume AML detection systems, this instability can reduce investigator trust [19]. Furthermore, Even with algorithmic improvements like TreeSHAP, computing SHAP values on datasets with millions of entries remains a significant computational bottleneck in industry-level pipelines, often taking many hours and undermining real-time usability. [20]

Despite its growing importance, XAI in AML faces three primary weaknesses: (i) instability: Different runs of the same model can yield varying explanations [21]; (ii) computational overhead: methods like SHAP can be prohibitively slow for large datasets [20] [22]; and (iii) limited adoption in AML: Most deployed systems remain black-box due to the perceived trade-off between interpretability and accuracy [21].

2.5.2 Critical Analysis of Social Network Analysis

Benchmarking on the work by Kim and Leskovec (2011). The study has shown that network completeness is critical: missing up to 50% of transaction edges can severely distort network metrics and impair detection accuracy. Additionally, scalability tests reveal that graph algorithms like community detection degrade sharply on graphs exceeding 50 million edges unless heavily optimized, which can be a barrier for large-scale payment networks [23].

The analytical power of SNA is constrained by data completeness, computational scalability, and susceptibility to misinterpretation of structural measures. In large financial transaction graphs, missing edges or misreported nodes can distort community detection and centrality rankings, leading to false confidence in detection results.

2.6 Related Work

2.6.1 Existing Approaches to Money Laundering Detection

2.6.1.1 Machine Learning Approaches

Several studies have demonstrated the potential and effectiveness of machine learning algorithms for AML, by comparing them with the traditional rule-based systems or other baseline methods. For instance, Harris et al [24] [7] developed a supervised learning model based on XGBoost, a popular gradient boosting algorithm, to prioritize which transactions should be manually investigated for potential money laundering. They used a large data set from Norway’s largest bank, DNB, and compared their model with the bank’s current alert system. They found that their model outperformed the existing system in terms of reducing false positives and increasing true positives. They also suggested that using non-reported alerts and normal transactions in the training data could improve the model performance.

Similarly, Harris et al [24] used various supervised learning techniques, such as logistic regression, LASSO, KNN, and XGBoost, to identify bank customers at an increased risk of committing money laundering. They used two proprietary data sets from Scotiabank, a large bank headquartered in Canada, and compared the model performance across different operationalizations of the outcome and the inclusion of novel features derived from the existing features. They found that logistic regression with a binary outcome and novel features performed the best, with an F1 score of 0.79 and accuracy of 0.72. They also recommended that investigators should consider different ways of defining the outcome and creating novel features to improve the detection of money laundering.

In addition to the individual machine learning algorithms, some studies have also explored the use of hybrid or ensemble methods, which combine multiple algorithms or models to achieve better results. For example, Lokanan et al [25] proposed a hybrid model that integrates machine learning and social network analysis to detect money laundering. They used a simulated data set of transactions that mimic the real-world money laundering scenarios and applied a two-stage approach. In the first stage, they used a supervised learning model based on random forest to classify the transactions as normal or suspicious. In the second

stage, they used a social network analysis technique based on centrality measures to rank the suspicious transactions according to their importance. They found that their hybrid model achieved a high accuracy of 0.98 and a low false positive rate of 0.02.

Another example of a hybrid method is the one proposed by Goldbarsht et al [26], who used a combination of unsupervised and supervised learning techniques to detect money laundering. They used a real data set of transactions from an Indian bank and applied a three-stage approach. In the first stage, they used an unsupervised learning technique based on k-means clustering to group the transactions into different clusters. In the second stage, they used a supervised learning technique based on decision tree to label the clusters as normal or suspicious. In the third stage, they used another supervised learning technique based on neural network to classify the transactions within the suspicious clusters as normal or suspicious. They found that their hybrid method achieved a high accuracy of 0.94 and a low false positive rate of 0.06.

The literature review shows that machine learning algorithms have a great potential to enhance the AML capabilities and performance, by leveraging the large and complex data sources in the financial sector, and by providing more accurate, efficient, and adaptive solutions. However, there are also some challenges and limitations that need to be addressed, such as:

The availability and quality of the data: Machine learning algorithms depend on the data to learn and make predictions, so the data quality and quantity are crucial for the model performance. However, obtaining and processing the data for AML can be difficult, due to the issues of privacy, security, confidentiality, and compliance. Moreover, the data may be noisy, incomplete, imbalanced, or outdated, which can affect the model accuracy and reliability [25].

The interpretability and explainability of the models: Machine learning algorithms, especially the complex and nonlinear ones, such as neural networks and deep learning, can be seen as black boxes, which means that their internal logic and reasoning are not transparent or understandable to the human users. This can pose a problem for the AML domain, where the models need to provide clear and convincing explanations for their decisions, especially when they flag a transaction or a customer as suspicious. Moreover, the models need to be

interpretable and explainable for the regulators and auditors, who need to verify and validate the compliance and effectiveness of the AML systems [25].

The ethical and legal implications of the models: Machine learning algorithms, especially the ones that use personal or sensitive data, such as customer profiles, behaviors, and preferences, can raise some ethical and legal concerns, such as privacy, fairness, accountability, and liability. For example, the models may violate the privacy rights of the customers, by collecting and using their data without their consent or knowledge. The models may also discriminate or harm some customers, by producing biased or erroneous results, based on their demographic or socio-economic characteristics. Furthermore, the models may create some accountability and liability issues, by shifting the responsibility and blame from the human actors to the machine agents [25].

2.6.1.2 Explainable Machine Learning Approaches

According to Shukla et al [27], advancements in computing and data generation have fueled research in machine learning over the past decade. While accurate predictions remain crucial, complex models often lack transparency and interpretability. In domains like medicine, banking, and law, where critical decisions impact human lives, the adoption of opaque models is challenging. Explainable machine learning has become essential for building trust and driving adoption in high-stakes areas like finance, crime, and healthcare. Regulations such as the General Data Protection Regulation (GDPR) mandate interpretability and explainability for AI/ML solutions.

A survey conducted by [28] emphasizes the literature on the importance of trust in the system and the genuine need for explainability, transparency, and interpretability for machine learning applications in judiciary, government, finance, and autonomous transport.

Explainable machine learning work is divided into two categories [29]: (i) transparency design that focuses on the internal workings of the model from the developer’s perspective, and (ii) post-hoc design, which explains why a result is inferred from the user’s perspective.

A systematic literature review conducted by Weber et al [30] highlighted the importance of explainable machine learning in finance. The study screened 2,022 articles from leading finance, information systems, and computer science outlets. The authors found that areas

like risk management, portfolio optimization, and applications around the stock market are well-researched, while anti-money laundering is understudied.

A critical review by Kute et al [31], discussed the application of deep learning (DL) techniques together with explainable machine learning for identifying suspicious money laundering transactions. The authors noted that 51% of machine learning methods used in anti-money laundering are non-interpretable, and explainable machine learning use is not seen in the anti-money laundering domain.

Researchers have proposed various methodologies to enhance the explainability of machine learning models in AML detection. These include feature importance techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), which provide insights into the contribution of individual features towards model predictions [32]. Additionally, rule-based systems and decision trees offer interpretable frameworks for understanding model decisions, albeit at the cost of reduced predictive performance [33]. Hybrid approaches that combine the strengths of black-box machine learning models with interpretable components are also being explored to achieve a balance between accuracy and explainability in AML detection [34].

2.6.2 Existing Machine Learning Systems that Detect Money Laundering

2.6.2.1 The FinCEN Artificial Intelligence System

The FinCEN Artificial Intelligence System (FAIS) is used to process and scrutinize all Bank Secrecy Act (BSA) reports, primarily focusing on Currency Transaction Reports (CTRs), which constitute over 90% of the received reports. The system operates with three fundamental categories of data: transactions, subjects, and accounts. Transactions associated with the same individual or business are amalgamated to create a distinct data element known as a subject, while transactions linked to the same bank account contribute to the formation of an element called an account. The classification of transactions involves scrutinizing transaction information, such as name, address, and social security number.

FinCEN’s knowledge-based system evaluates these three types of data elements, based on indicators directly embedded in transactions and additional indicators derived from them. It

assigns a numeric suspiciousness score to each transaction, subject, and account, considering these scores along with other criteria. FinCEN analysts then choose subjects and accounts for further scrutiny. The link analysis component of FAIS helps identify networks of financial activities, distinguishing between legitimate business activities and potential money laundering.

FAIS has been operational since March 1993, processing around 200,000 transactions weekly. By January 1995, the system had processed 20 million transactions, resulting in 3 million consolidated subjects and 2.5 million accounts. As of May 1995, FAIS generated over 400 investigative support reports corresponding to over 1 billion USD in potentially laundered funds [35].

2.6.2.2 Fedwire Scanning System

The Federal Reserve Bank of America has the ability to electronically scan and retrieve records of wire transfers through its Fedwire system. Law enforcement agencies can request a search of Fedwire records with an appropriate search warrant, which can specify up to twenty different character strings representing distinct items such as names, account numbers, street addresses, and spellings of names. However, these searches have limitations, such as being limited to records from the past 180 days, reviewing records from one Reserve Bank's Fedwire traffic, and taking approximately one hour to search a single day.

Despite these constraints, law enforcement agencies may opt for Fedwire scanning over relying on records from individual banks. The identity of the bank involved in the transfer may not be known among the 11,700 banks with Fedwire access, and if suspicions arise about money laundering or bank employees' involvement, law enforcement may prefer to secure the records through Fedwire scanning [35].

2.6.2.3 Electronic Fraud Detection (EFD) system

The Electronic Fraud Detection (EFD) system is a crucial tool for fraud investigators at The Travelers Insurance Companies to detect and analyze healthcare provider fraud. Traditionally, fraud detection relied on manual inspection of claim forms and tips from various sources. However, the increasing prevalence of electronic records has enabled automated

analysis, eliminating traditional methods like scrutinizing paper claim forms for signs of alteration. The EFD system was developed to identify fraud through automated analysis, creating a detailed statistical profile for each healthcare provider and comparing it to similar profiles. The system generates statistics indicating the frequency of specific types of services and the occurrence of unexpected services by a particular provider. Potential fraud cases are flagged when a provider deviates from statistical norms in statistically significant and indicative ways. The EFD system employs heuristics to explain why a particular statistic suggests fraud and what types of deviations from the average are considered significant. [36].

The literature review shows that the Anti Money Laundering domain being confidential in nature—Many researched methodologies for detecting suspicious money laundering transactions are patented and hence they are not published for obvious reasons. Therefore such innovative technologies may be applicable and beneficial to the African financial sector, especially in terms of improving money laundering detection.

2.6.3 The Role of Social Network Analysis in AML Detection

Social Network Analysis (SNA) has emerged as a powerful tool for identifying and analyzing complex networks of financial transactions associated with money laundering activities. By modeling the relationships and interactions between entities such as individuals, businesses, and financial institutions, SNA enables the detection of suspicious patterns and anomalous behaviors indicative of illicit financial activities [37]. Furthermore, SNA provides valuable insights into the structure and dynamics of illicit networks, facilitating the identification of key nodes, pathways, and laundering techniques employed by criminals [38].

Network analysis enables us to measure and visualize the characteristics and patterns of money laundering networks, such as the centrality, density, clustering, and community detection of the nodes and edges [39]. Combining these algorithms with a machine learning model can help improve the efficiency of detecting Money laundering.

While SNA offers valuable insights into the structure of illicit networks, the increasing adoption of machine learning techniques for AML detection has raised concerns regarding the transparency and interpretability of model decisions. Black-box machine learning models, such as deep neural networks, often lack transparency, making it difficult to understand the

rationale behind their predictions. In the context of AML, where regulatory compliance and accountability are paramount, the integration of explainable machine learning techniques becomes imperative to justify model decisions and enhance trust in automated detection systems [40].

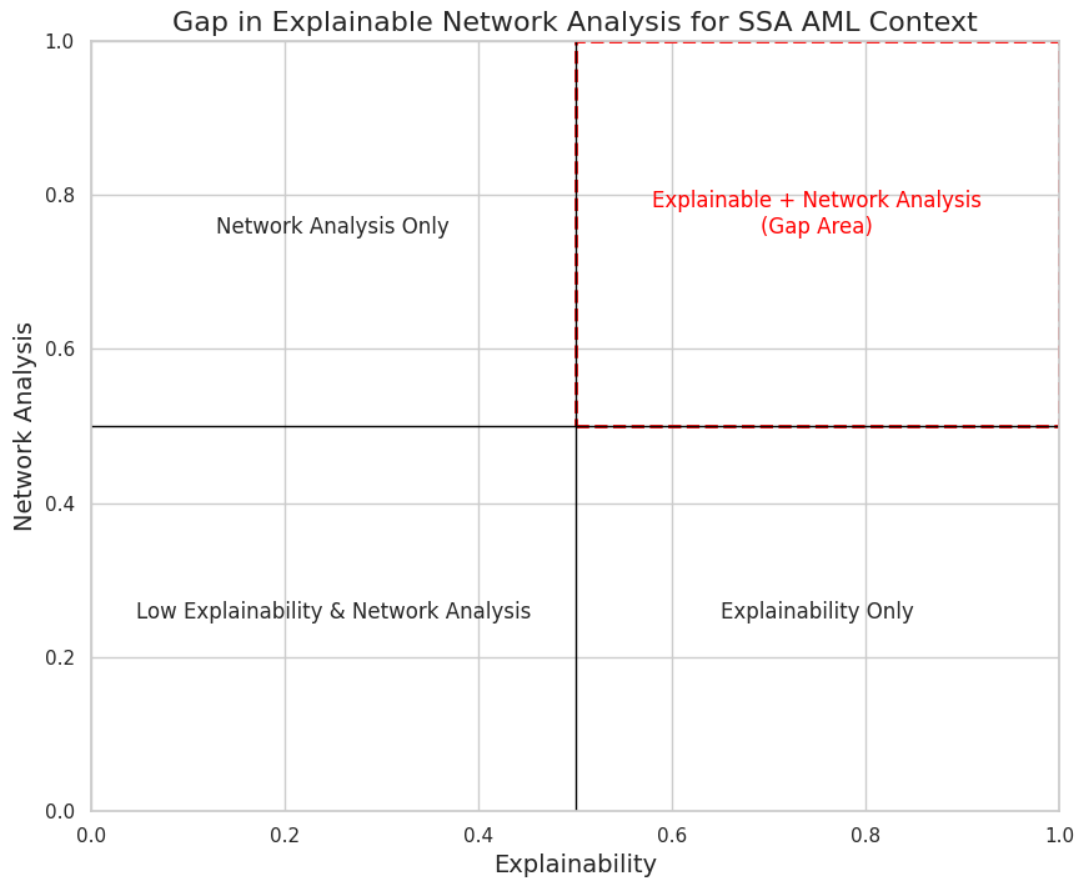
This research aims at integrating Explainable Machine Learning techniques with SNA approaches to enhance the transparency and interpretability of AML detection models. Explainable machine learning methods such as SHAP and LIME will be applied to SNA-based models to provide insights into the contribution of individual nodes and edges towards model predictions. By visualizing the importance of network features and highlighting suspicious nodes or transactions, the explainable machine learning-SNA hybrid model will enable investigators and analysts to understand and trust the output of AML detection systems.

2.6.4 Contextualizing the Research Gap

Despite the growing recognition of the importance of explainable machine learning models and social network analysis (SNA) in anti-money laundering (AML) detection, there exists a significant research gap in understanding the synergistic integration of these two methodologies. While individual studies have explored the application of Explainable Machine Learning techniques to enhance the interpretability of AML detection models and the utility of SNA in uncovering complex financial networks, there is limited research that investigates how Explainable Machine Learning can be effectively combined with SNA to provide transparent and actionable insights into illicit financial activities. Thus, a research gap exists in developing and evaluating novel Explainable Machine Learning-SNA hybrid models that leverage the strengths of both approaches to improve the transparency, interpretability, and effectiveness of AML detection systems. The literature also indicates that while certain areas like risk management are well-researched, anti-money laundering remains an area needing more focus. Addressing this research gap is crucial for advancing the state-of-the-art in AML detection and enhancing the ability of financial institutions and regulatory agencies to combat money laundering and illicit financial activities effectively.

2.6.5 Research Gap Visualisation

The diagram below illustrates the positioning of this research relative to existing works. Most prior studies focus on either Explainable AI without network analysis, or SNA without explainability. Hybrid approaches exist but are either non-explainable or not domain-tuned for Sub-Saharan financial data.



2.7 Conclusion

The examination of the existing literature has underscored the potential of integrating explainable machine learning models and Social Network Analysis (SNA) methodologies in the realm of Anti Money Laundering (AML) detection. The synthesis of existing research underscores the imperative to not only identify but also thwart money laundering activities within financial institutions. It has revealed the challenges faced by conventional systems and emphasized the necessity to evolve current methodologies to keep pace with sophisticated money laundering tactics.

Studies, such as the work by Lokanan et al. [25], have demonstrated the efficacy of combining machine learning with network analysis to achieve high accuracy and low false positives in detecting money laundering patterns. Hence, this review advocates for the development and implementation of explainable machine learning models integrated with SNA techniques. Such an approach would enable the interpretation of specific relationships, flows, and patterns within money laundering networks, thereby enhancing detection accuracy and providing actionable insights tailored to the intricacies of financial transactions and hidden connections inherent in illicit financial activities.

Moving forward, it is imperative to develop and deploy hybrid machine learning models that ensure transparency and interpretability in AML detection. Techniques such as network analysis, when combined with model explanation, will bolster the trustworthiness of AML detection systems. Additionally, involving human experts and stakeholders in the model development and evaluation process will provide invaluable feedback, guidance, and validation, ensuring the practical utility and effectiveness of the proposed methodologies.

In essence, the integration of explainable machine learning and Social Network Analysis in AML detection holds promise for advancing the effectiveness and efficiency of financial crime detection processes. By addressing the limitations of existing techniques and leveraging the strengths of explainable machine learning and SNA, will diminish the prevalence of inaccurate outcomes and covering gaps of inadequate money laundering technologies in Africa.

3 Methodology

This study will implement a hybrid supervised learning and network analysis framework. While the panel suggested exploring unsupervised learning for its potential in uncovering entirely novel patterns, the decision to proceed with supervised learning is informed by the presence of labeled Interswitch transaction datasets, the necessity for explainable outputs, and alignment with regulatory compliance.

Explainability will be integrated at two stages: (i) feature attribution for both raw transaction features and network-derived features using SHAP; and (ii) network visualization overlays that highlight high-importance nodes and edges for investigators. This dual approach ensures both model-level and structural transparency.

3.1 Studying existing financial systems within Sub-Saharan Africa (Uganda as a case study)

Most of the Financial Technology institutions in Uganda have a core banking or wallet system that handles the transactions. Different patterns are used to identify potential money laundering transactions. The bank’s core/wallet system is used to monitor and identify transactions that meet the configured patterns such as.

- (i) Transactions amounts that are above a certain threshold [8].
- (ii) Transactions in tranches that occur in small amounts and the frequency is higher than the configured threshold.
- (iii) Deposits and withdrawals that are done on the same day [8].
- (iv) Near border transactions i.e. to or from the neighboring countries [8].
- (v) Transactions by customers profiled have high risk such as Forex Business [8].
- (vi) Wire transfers from and to countries viewed as high risk (War zone areas) [8].

A local fin-tech will be used as a case study and the information gathered will help us understand the patterns that are used to identify money laundering transactions and

identify loopholes that criminals use to avoid detection. Some loopholes have been closed in the implemented system.

3.2 Data Collection and Feature Engineering

3.2.1 Questionnaires and Interviews

Data collection from Agency banking shops and Banks and SACCOs using questionnaires and Interviews Data collection will enable us:

- (i) To provide first-hand information from the stakeholders involved in the financial sector, such as bank managers, agents, customers, and regulators.
- (ii) To understand the current practices and challenges of detecting and preventing money laundering in Sub-Saharan Africa.
- (iii) To identify and select the most relevant features and data sources for training machine learning models for money laundering detection, considering the informal economy and cash-based transactions.

3.2.2 Dataset Sources

Gather financial transaction data from various sources, including financial institutions. (InterSwitch Group) and publicly available money laundering dataset (Kaggle), data Augmentation.

InterSwitch Group Dataset: This dataset will be used as both training and validation dataset (Domain: Card Transactions and transactions through Agency Network).

3.2.2.1 Agent Transactions Dataset

Size: Contains 30,506 wallet accounts.

Number of Transactions: Includes 323,329 transactions.

Type of Data: Transaction records involving wallet accounts.

Structure: Tabular data with various attributes (e.g., Id, Amount, CustomerId, TransactionCode, and so on.).

Temporal Properties: Transaction data collected over time.

Source: Transactions conducted through agents.

3.2.2.2 ATM Transactions Dataset

Size: Comprises 500,000 ATM transactions.

Type of Data: ATM transaction records.

Structure: Tabular data with various attributes (e.g., Id, Amount, card number, Transaction-Code, and so on.).

Temporal Properties: Time-stamped data.

Source: ATM terminals.

3.2.2.3 Synthetic Data Dataset

Generate synthetic data generation based on Local data set (Interswitch Group Dataset). Due to lack of available labeled dataset we'll use synthetic data generation to produce large volumes of synthetically generated data (labelled) [41]

3.2.3 Feature Engineering

Create relevant features such as transaction frequency, size, and network characteristics. Incorporate additional data, such as geopolitical events and market fluctuations.

3.3 Implementation

3.3.1 Model Selection and Training

Algorithm Selection: Choose machine learning algorithms tailored to the SSA context.

- (1) Logistic Regression
- (2) Tree Based Algorithms - Decision Tree and Random Forest
- (3) Support Vector Machines.

Algorithm	Hyper paramters
Logistic Regression	Default
Random Forest	Number of trees in the Forest = 100, Default
Support Vector Machine	Linear Kernel, C= 1, Default
Decision Tree	Default
Mullti-Layer Perceptron	Learning Rate = 0.01, batch size=100, default

Table 2: Fine turning of Algorithm Hyper parameters will be done to improve the accuracy of the results

- (4) Multi-Layer Perceptron Algorithm.
- (5) All the above will be combined with network analysis and graph theory to improve their efficiency.

3.3.2 Model Evaluation

The evaluation metrics used for analyzing the performance of the algorithms are commonly used when the data is highly distributed [42], which is the case with the Interswitch Group dataset.

The following metrics will be considered during model evaluation.

3.3.2.1 Accuracy

The model's accuracy will be based on the percentage of correct predictions out of all predictions. Accuracy is a simple and intuitive metric, but it can be misleading for imbalanced datasets, where one class is much more frequent than the others. For example, if 90% of the data belongs to class A, and 10% to class B, a model that always predicts class A will have 90% accuracy, but it will not be useful for classifying classes.

3.3.2.2 Confusion matrix

This will be represented as a table that summarizes the prediction results of a classification model, by showing the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) for each class. A confusion matrix can help to identify the strengths and weaknesses of a model, such as which classes are confused with each other, or which classes have more false positives or false negatives.

3.3.2.3 Precision

The ratio of true positives to the total number of positive predictions. Precision measures how accurate the model is when it predicts a positive class. A high precision means that the model has a low rate of false positives, and it is reliable when it predicts a positive class.

3.3.2.4 Recall

This will be represented as the ratio of true positives to the total number of actual positives. Recall measures how complete the model is when it predicts a positive class. A high recall means that the model has a low rate of false negatives, and it can capture most of the actual

positives1 F1 score: the harmonic mean of precision and recall. F1 score is a single metric that combines both precision and recall, and it balances between them. A high F1 score means that the model has both high precision and high recall, and it can accurately and completely predict the positive class.

3.3.2.5 ROC curve and AUC

This will be a graphical representation of the trade-off between the true positive rate (TPR) and the false positive rate (FPR) of a model, as the decision threshold varies. The ROC curve plots the TPR (which is the same as recall) against the FPR (which is the inverse of precision) for different values of the threshold. The AUC (area under the curve) is the area under the ROC curve, and it measures the overall performance of the model across all possible thresholds. A high AUC means that the model can distinguish well between the positive and negative classes, regardless of the threshold.

3.3.2.6 Precision-recall curve

This will be a graphical representation of the trade-off between precision and recall of a model, as the decision threshold varies.

The precision-recall curve plots precision against recall for different values of the threshold. The precision-recall curve is useful for imbalanced datasets, where the ROC curve can be misleading. A high area under the precision-recall curve means that the model can achieve both high precision and high recall, regardless of the threshold.

3.3.2.7 Log loss or cross entropy loss

A measure of the uncertainty or error of the model's probability estimates for the classes. Log loss penalizes the model for assigning a low probability to the correct class, or a high probability to the incorrect class. A low log loss means that the model is confident and accurate in its predictions.

3.3.3 Labeling

Annotate transactions as “normal” or “suspicious” using historical data, expert knowledge, or emerging patterns.

3.3.4 Training and Validation

Split data into training, validation, and test sets. Train and fine-tune models while considering the region’s specific money laundering indicators.

3.3.5 Enhanced Evaluation Strategy

In addition to traditional ML metrics (Accuracy, F1, ROC-AUC), domain-specific KPIs will be tracked: (i) reduction in false positives compared to baseline bank AML systems; (ii) investigator time saved per alert; (iii) detection of previously unflagged laundering cases validated by compliance teams. An ablation study will measure the individual contribution of XAI components and SNA features to overall model performance.

3.3.6 Real-world Deployment and Scalability

Deploy the trained model in financial institutions, regulatory agencies, and other relevant stakeholders’ infrastructure for real-time or batch processing. The solution will be designed for modular deployment as a containerized microservice. It will expose a secure REST API for integration with core banking systems, financial processors and regulatory reporting platforms. Batch processing will be used for historical analysis, while a streaming architecture (Apache Kafka + Spark Streaming) will enable near-real-time detection for high-value transactions. Scalability will be ensured through horizontal scaling ”and distributed graph computation frameworks.

3.4 Proposed Draft Framework For Detecting Money Laundering

Input Dataset: Account Id (source and Destination id), Amount, Transaction type, Transaction Id and Time Stamp

Using a Hybrid Machine learning model(Combined with Network Analysis) to classify transactions.

Reporting(Interpretable Model).

3.5 Mapping of Specific Objectives to Method

This subsection provides a summary of the mapping of specific objectives of the research study to the methods that will be used.

Specific Objective	Method
Identify and characterize properties of common money laundering scenarios	- Studying existing financial systems within Sub-Saharan Africa. - Questionnaires and Interviews with stakeholders.
Identify and select the most relevant features and data sources for training the explainable machine learning model and social network analysis for money laundering detection	Obtain dataset and carry out data exploration to ensure the selected features are indicative of money laundering activities within the specified context.
Develop and evaluate an explainable machine learning model with network analysis capable of identifying anomalous patterns and potential money laundering activities.	- Select appropriate Models that are to be used for detection of money laundering patterns. - Train and fine-tune machine learning models using the identified features and data sources to optimize performance for money laundering detection. - Label datasets by annotating transactions as “normal” or “suspicious” using historical data, expert knowledge, or emerging patterns . - Evaluate the developed models through rigorous testing and validation procedures, considering the intricacies of the Sub-Saharan African financial landscape.

Table 3: Summary of specific objectives mapped to methods.

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Appendices

Appendix A : Work Plan

